MIL-C-12436C
8 April 1971
SUPERSEDING
MIL-C-12436B
8 April 1968

MILITARY SPECIFICATION

CROSSARMS, WOOD

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers treated wood crossarms.

1.2 Classification. Crossarms shall be of the following types and classes as specified:

Type I - Electric or Heavy Telephone Arms - 3-1/4 x 4-1/4
Type II - Pony Telephone Arms - 2-3/4 x 3-3/4
Type III - National Electric Light Association Arms (Light Arms) - 3-1/4 x 4-1/4
Type IV - New England Arms - 3-1/4 x 4-1/4
Type V - Transmission Arms - 3-3/4 x 4-3/4
Type VI - REA Telephone Arms - 3-3/4 x 4-3/4
Type VII - REA Distribution Arms
Class 1 - 3-1/2 x 4-1/2
Class 2 - 3-3/4 x 4-3/4

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-L-14362 - Lumber, Unitizing and Loading of.

FSC 5510

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STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1363	- Measurement of Wood Moisture Content.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

EDISON ELECTRIC INSTITUTE

TD-90 - Specification for Douglas Fir Crossarms, Treated or Untreated.
TD-91 - Specification for Dense Southern Pine Crossarms, Preservative Treated.

(Application for copies should be addressed to the Edison Electric Institute, 750 Third Avenue, New York, N. Y. 10017.)

AMERICAN WOOD-PRESERVERS' ASSOCIATION (AWPA)

C1 - All Timber Products-Preservative Treatment by Pressure Processes.
C25 - Sawn Crossarms-Preservative Treatment by Pressure Processes.
C26 - Standard for the Preservative Treatment of Crossarms by Non-Pressure Processes, Douglas Fir, Western Larch and Western Hemlock.
P1 - Standard for Coal Tar Creosote for Land and Fresh Water Use.
P8 - Standards for Oil-Borne Preservatives.
P9 - Standard for Hydrocarbon Solvents for Oil-Borne Preservatives.
M2 - Standard for Inspection of Treated Timber Products.
M6 - Brands Used on Forest Products.

(Application for copies should be addressed to the American Wood-Preservers' Association, 1012 14th Street NW, Washington, D. C. 20005.)

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3. REQUIREMENTS

3.1 Description. Crossarms shall be made of either Douglas Fir or Dense Southern Pine.

3.2 Material. Material shall be as specified.

3.2.1 Douglas Fir. Douglas Fir shall conform to Edison Electric Institute, TD-90.

3.2.2 Dense Southern Pine. Dense Southern Pine shall conform to Edison Electric Institute, TD-91.

3.3 Dimension. Unless otherwise specified, crossarms shall conform to table I. Length, boring details, pin hole arrangement, and brace bolt hole spacing shall be selected from table I. Tolerances shall be those listed in Appendix I of TD-90 or TD-91 as applicable.

3.3.1 Blank crossarms. When specified, blank crossarms of the cross section and length specified shall be furnished and shall be treated or untreated as specified.

3.4 Pin hole spacing. Center and side pin hole spacing shall be measured center of hole to center of hole; end pin hole spacing shall be measured center of hole to end of crossarm.

3.5 Brace bolt holes. Unless otherwise specified, all crossarms shall be bored for two brace bolt holes spaced as specified in table I (hole size 7/16-inch).

3.6 Incising. Unless otherwise specified, Douglas Fir crossarms shall be incised on all faces to a depth not less than 3/16 inch and not to exceed 1/4 inch in accordance with TD-90.

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Table I. Dimensions

Length of Arm Feet	Length of Arm Inches	No. of Pins	Pin Holes Spacing (Inches) Center	Sides	Ends	Size Inches	Diameter of Center Bolt Hole Inches	Brace Bolt Spacing Inches	Use Length Brace Inches
Type I		3-1/4 X 4-1/4 in. Electric Arms or Heavy Telephone Arms							
3	-	2	28	-	4	9/16	11/16	25	20
4	-	4	16	12	4	1-17/32	11/16	28	22
5	-	4	18	17	4	9/16	11/16	28	22
6	-	4	22	21	4	9/16	11/16	32	24-26
6	-	6	16	12	4	9/16	11/16	32	24-26
8	-	6	18	17-1/2	4	9/16	11/16	32	24-26
8	-	8	16	12	4	1-17/32	11/16	32	24-26
8	6	10	16	9-3/4	4	9/16	11/16	32	24-26
8	6	10	16	9-3/4	4	9/16	11/16	28	22
10	-	8	17-1/2	15-3/4	4	*	11/16	42	30-32
10	-	10	16	12	4	1-17/32	11/16	42	30-32
10	-	10	32	10	4	9/16	11/16	42	30-32
10	-	12	16	9-5/8	3-7/8	*	11/16	42	30-32
10	-	4	9-1/2	-	3-1/2	11/16	11/16	-	-
10	-	18	16	**	3	1-17/32	11/16	43	-
10	-	18	16	**	3	1-9/32	11/16	43	-
10	-	Blank Crossarm							
Type II		2-3/4 X 3-3/4 in. Pony Telephone Arms							
3	6	4	16	9-1/2	3-1/2	9/16	11/16	28	22
3	6	4	16	9-1/2	3-1/2	1-9/32	11/16	28	22
5	2	6	16	9-3/4	3-1/2	1-9/32	11/16	28	22
6	10	8	16	9-3/4	3-3/4	9/16	11/16	28	22
10	-	12	16	9-5/8	3-7/8	9/16	11/16	28	22
Type III		3-1/4 X 4-1/4 in. National Electric Light Association (light arms)							
3	2	2	30	-	4	9/16	11/16	28	22
5	7	4	30	14-1/2	4	*	11/16	38	28
8	-	6	30	14-1/2	4	11/16	11/16	38	28
8	-	6	30	14-1/2	4	1-17/32	11/16	38	28
9	2	8	30	12	4	*	11/16	38	28
11	-	10	28	12	4	1-17/32	11/16	42	-
Type IV		3-1/4 X 4-1/4 in. New England Arms							
3	-	2	30	-	3	9/16	11/16	33	26
5	6	4	30	13-1/2	4-1/2	*	11/16	36	28
7	9	6	30	13-1/2	4-1/2	*	11/16	36	28
10	-	8	30	13-1/2	4-1/2	*	11/16	36	28
Type V		3-3/4 X 4-3/4 in. Transmission Arms							
10	-	8	30	13-5/8	4-1/8	*	13/16	44	-
10	-	4	38	36	5	*	13/16	60	-
Type VI		3-3/4 X 4-1/4 in. REA Telephone Arms							
1	6	2	12	-	3	*	11/16	16	-
6	-	6	16	12	4	*	11/16	25	-
6	8	6	32	10	4	*	11/16	25	-
Type VII, Class 1		3-1/2 X 4-1/2 in. REA Distribution Arms							
4	-	2	40	-	4	*	11/16	24	-
5	7	4	30	14-1/2	4	*	11/16	38	-
5	7	4	30	14-1/2	4	1-17/32	11/16	42	-
8	-	6	30	14-1/2	4	*	11/16	38	-
8	-	8	16	12	4	1-17/32	11/16	32	-
Type VII, Class 2		3-3/4 X 4-3/4 in. REA Distribution Arms							
8	-	4	30	-	3	*	11/16	-	-
10	-	6	38	18-1/2	4	*	11/16	-	-

*Pin holes, other than above, may be specified in accordance with the following:

Pin hole diameters in relation to pin size:

1-1/4 in. wood pins, specify pin hole diameter 1-9/32 inches.
1-1/2 in. wood pins, specify pin hole diameter 1-17/32 inches.
1/2 in. steel pins, specify pin hole diameter 9/16 inch.
5/8 in. steel pins, specify pin hole diameter 11/16 inch.
3/4 in. steel pins, specify pin hole diameter 13/16 inch.

**For pin hole spacing, see figure 1.

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3.7 Preservative treatment. Unless otherwise specified, all crossarms shall be preservative treated with either of the preservatives listed in table II.

Table II. Treatment of Wood Crossarms

Species	Net retention: Creosote (AWPA Std. P1) lbs. per cu. ft.*	Net retention: Pentachlorophenol (AWPA Std. P8 and P9) lbs. per cu. ft.*	Treating Specification (AWPA Standards)
Southern Pine	8.0	0.40	C1, C25
	10.0	0.50	C1, C25
Douglas Fir	6.0	0.30	C1, C25, C26
	8.0	0.40	C1, C25, C26

*Net retention values given are minimum acceptable value for either treatment.

3.7.1 Retention. Unless otherwise specified, the lower minimum net retention for each solution and species shall be as specified in table II.

3.7.2 Preservative. The preservative used shall conform to the applicable AWPA standard when tested in accordance with 4.4.2.3.

3.8 Edges. The edges shall be rounded or chamfered as specified in TD-90 and TD-91, except that when specified the top center shall not be chamfered or rounded.

3.9 Grades. Douglas Fir and Southern Pine shall conform to the grade limitations of TD-90 or TD-91 as applicable. Unless otherwise specified, Douglas Fir shall be furnished in either dense or close grain grade or a mixture of both grades.

3.10 Identification marking. Each crossarm shall be marked with typical brand and key information contained in AWPA M6. The information shall be burned-branded legible and permanently on the crossarm or shall be included on a recessed noncorrosive metal tag attached thereto. The code letters shall be not less than 1/2 inch high if burn-branded, and not less than 1/8 inch high if on a metal tag; metal tags shall be attached after treatment. Typical brand as follows:

ABCO = Suppliers brand
71 = Year of treatment
SPC = Species of timber and preservative treatment
8 = Retention pounds per cubic foot.

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3.11 Seasoning. Lumber used in the manufacture of the crossarms shall be kiln dried or air dried to moisture content of 22 percent or less. Moisture content shall be determined prior to treatment as specified in 4.4.2.1.

3.12 Penetration.

3.12.1 Depth of penetration. Depth of penetration of preservative in sap wood shall be not less than 100 percent.

3.12.2 Longitudinal penetration. Longitudinal penetration of preservative from ends, pin holes, and bolt holes shall be as follows:

(a) Douglas Fir - 2, 2-1/2 or 3 inches as specified.
(b) Southern Pine - 1-1/2 inches, minimum.

3.13 Workmanship. All crossarms shall be dressed on four sides, although hit and miss dressing on the bottom and slight skips on one side are permissible. Splintering around holes resulting from bits breaking through shall be removed. Prior to treatment, crossarms shall be free of dirt or other extraneous substances detrimental to treatment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Unit of product. For the purpose of inspection, each crossarm shall be considered a unit of product.

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4.3.2 Lot. All crossarms in the same charge shall be considered a lot.

4.3.3 Sampling.

4.3.3.1 Sampling of crossarms. Sampling of crossarms for examination and test shall be in accordance with MIL-STD-105.

4.3.3.2 Sampling of preservative. During treatment of crossarms, the preservative shall be sampled as specified in AWPA Standard M2.

4.3.4 Samples.

4.3.4.1 Samples of crossarms. Samples of crossarms selected in accordance with 4.3.3.2 shall be examined and tested as specified in 4.3.5.2.2.

4.3.4.2 Samples of preservative. Samples of preservative selected in accordance with 4.3.3.2 shall be tested as specified in 4.3.5.2.2.

4.3.5 Examination and test.

4.3.5.1 Examination. Samples selected in accordance with 4.3.3.1 shall be examined as specified in 4.4.1. AQL shall be 2.5 percent defective.

4.3.5.2 Tests.

4.3.5.2.1 Samples. Samples selected in accordance with 4.3.3.1 shall be tested as specified in 4.4.2.1, 4.4.2.2, and 4.4.2.3.

4.3.5.2.2 Preservative. During treatment of the crossarms the preservative shall be tested as specified in 4.4.2.4. Failure of the test shall be cause for rejection.

4.3.5.2.3 Retention of preservative. Samples selected in accordance with 4.3.3 shall be tested as specified in 4.4.2.2 for net retention of preservative. Rejected charges shall be subject to one retreatment only.

4.4 Inspection procedure.

4.4.1 Examination.

4.4.1.1 Examination prior to treatment. Prior to treatment the crossarms shall be examined for quality of lumber and for conformance to the specified construction detail. Presence of one or more defects shall be cause for rejection of the charge.

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4.4.1.1.1 Quality of lumber. The charge shall be examined for the defects specified in TD-90 and TD-91 as applicable.

4.4.1.1.2 Construction detail. The charge shall be examined for the following defects:

101. Species not as specified.
102. Type or type and class not as specified.
103. Dimensions of crossarms not as specified.
104. Brace holes and pin holes not as specified.
105. Not incised as specified.
106. Marking missing, illegible, or not as specified.
107. Cleanliness of crossarms not as specified.
108. Workmanship not as specified.

4.4.1.2 Examination after treatment. After treatment each charge shall be examined for the following defects. Presence of one or more defects shall be cause for rejection of the charge.

109. Mechanical damage to individual crossarms.
110. Preservative treatment not as specified.
111. Injury or deformation of the wood caused by excessive pressure.
112. Damage to wood caused by improper treating methods.

4.4.2 Tests.

4.4.2.1 Moisture content. Moisture content shall be determined in accordance with MIL-STD-1363. When the moisture content is determined by electrical resistance type moisture meter, the meter shall be equipped with insulated electrode needles. The needles shall be driven to depth of approximately one-fourth of the thickness of the piece. Failure of the charge to meet the specified average moisture content shall be cause for rejection of the charge.

4.4.2.2 Net retention of preservative. Net retention of preservative shall be determined in accordance with AWPA Standard M2, method as follows:

(a) Retention by extraction assay.
(b) Retention by chemical assay.

Either of the above methods may be used at option of the supplier to determine the retention. Nonconformance to 3.7.1 shall constitute failure of this test.

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4.4.2.3 Depth of penetration. Depth of penetration shall be determined in accordance with AWPA Standard M2. Nonconformance to 3.11 shall constitute failure of this test.

4.4.2.4 Preservative analysis. During treatment of the crossarms, the preservative shall be analyzed for conformance to the applicable AWPA preservative standard listed in table II as specified in AWPA Standard M2. Nonconformance to 3.7.2 shall constitute failure of this test.

4.5 Inspection of preparation for delivery.

4.5.1 Packing. The unitizing and loading of crossarms shall be inspected to determine compliance with MIL-L-14362.

4.5.2 Marking. The marking of crossarms shall be inspected to determine compliance with MIL-STD-129, or as otherwise specified.

5. PREPARATION FOR DELIVERY

5.1 Packing. Crossarms shall be unitized and loaded in accordance with MIL-L-14362 as specified.

5.2 Marking. Marking shall be in accordance with MIL-STD-129 unless otherwise specified.

6. NOTES

6.1 Intended use. The crossarms are intended to be used for support and to space the wires on wooden telephone and electric poles.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Types and classes required (see 1.2).
(c) Material required (see 3.2 and 6.4).
(d) Dimension required if other than as specified in table I (see 3.3 and 6.3).
(e) When blank crossarms are required and dimension and treatment desired, when applicable (see 3.3.1).
(f) Size and spacing of brace bolt holes required when other than those listed in table I (see 3.5).
(g) When incising of Douglas Fir crossarms is not required (see 3.6).

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(h) Preservative treatment required when other than as specified (see 3.7).
(i) Net retention required when other than as specified (see 3.7.1).
(j) When top center need not be chamfered (see 3.8).
(k) Whether close grain, dense grain, or combination grade of Douglas Fir is required (see 3.9).
(l) Penetration required for Douglas Fir (see 3.12.2 (a)).
(m) Packing and marking required (see 5.1 and 5.2).

6.3 Crossarms listed in table I. Table I lists crossarms normally procured for military use. If special boring details, such as, bolt spacing, pin or bolt sizes are required by the ordering activity and these features are not listed in table I, the ordering activity should furnish the contracting officer with the necessary engineering data. The boring details should be detailed on a drawing similar to figure 1.

6.4 Selection of material. At the time of procurement, the contracting officer should determine whether Douglas Fir or Dense Southern Pine is the most economical, based upon the location where delivery is to be made.

Custodians:

Army - ME
Navy - YD
Air Force - 84

Preparing activity:

Army - ME

Project No. 5510-0102

Review activities:

Army - CE, EL

User activity:

Navy - MC

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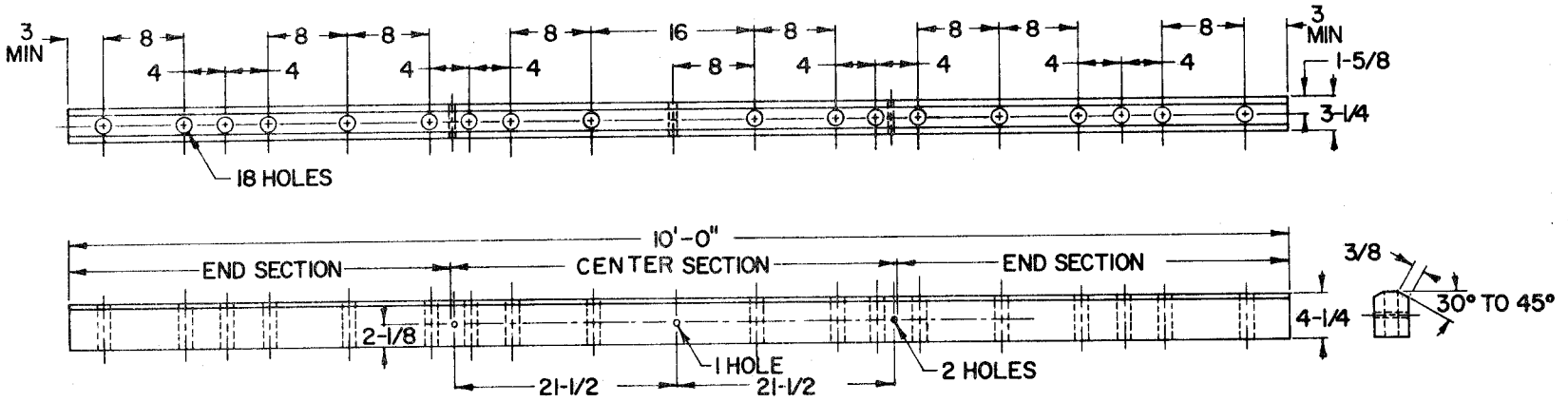


Figure 1. PIN HOLE SPACING FOR 18 PIN CROSSARM

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-12436C Crossarms, Wood

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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